

Sub-GeV Direct Dark Matter Detection: Improving Phonon Collection Efficiency in Phonon-Mediated Kinetic Inductance Detectors Using Simulations and Analysis

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0

**Background***Dark Matter and Kinetic Inductance Detectors*

1

**Quasiparticle Trapping Fins***Sonnet Simulations and Observations*

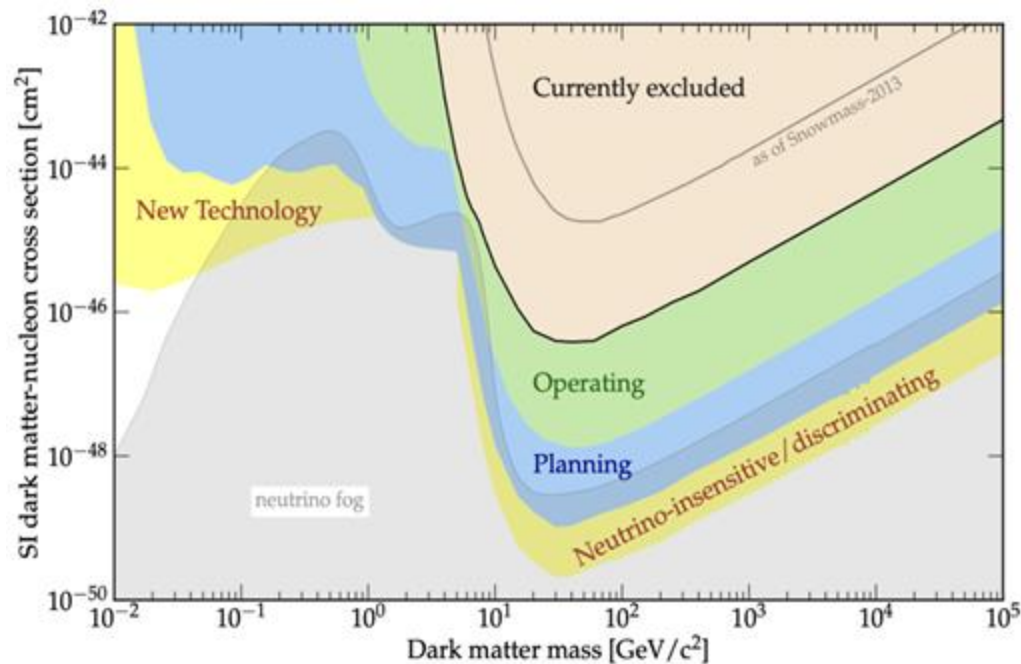
2

**Detector Mounting Box***Wire Bonding and Design Iterations*

3

**Future Work**

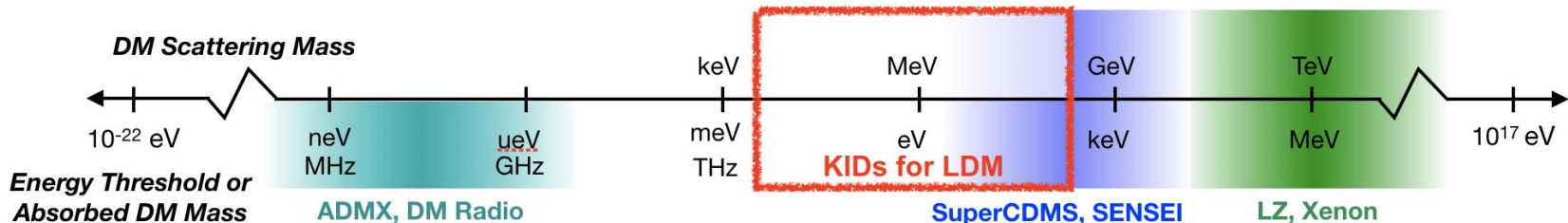
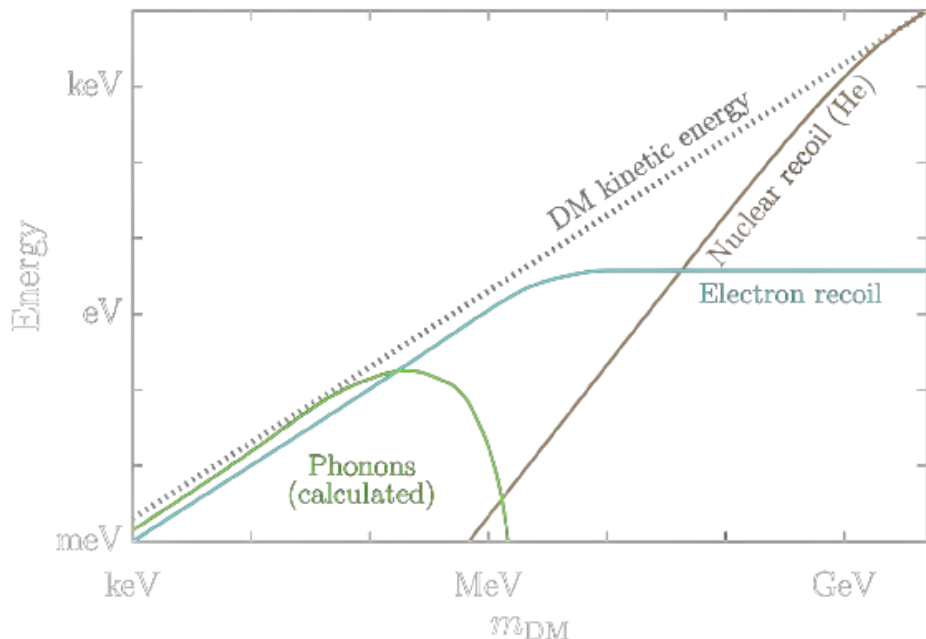
0. Background



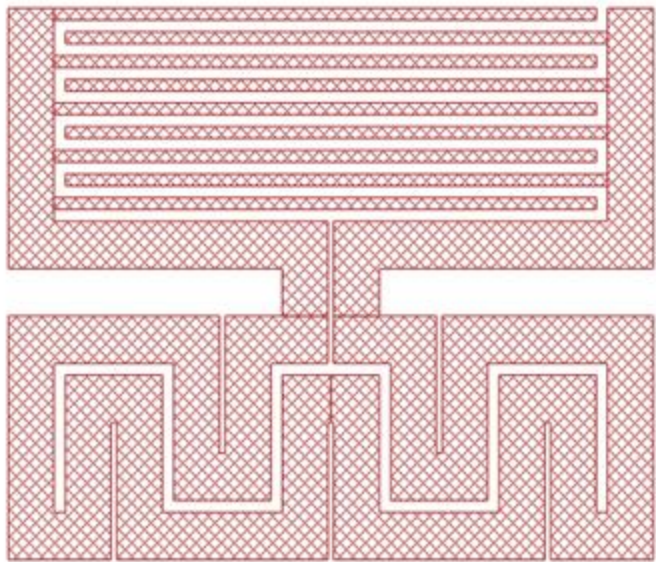
Snowmass 2021 CF1 (Particle DM) white paper, arXiv:2209.07426

- Lots of candidates for dark matter (WIMPs: GeV-TeV, QCD Axions: 10^{-5} eV – 10^5 eV)
- We examine the sub-GeV mass range of dark matter ('New Technology' section)
- Requires ultrasensitive phonon mediated detectors
- Energy deposition is too small to easily measure with scintillation or ionization

- As mass decreases, energy deposited decreases, phonons become the dominant energy-carriers
- Thus, we look at phonon-mediated detectors

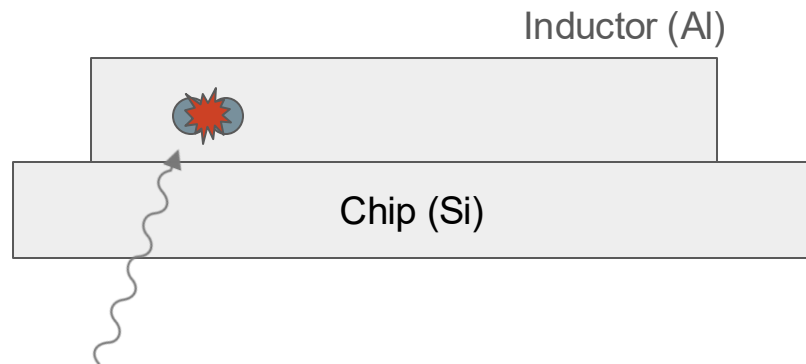


0. Kinetic Inductance Detector

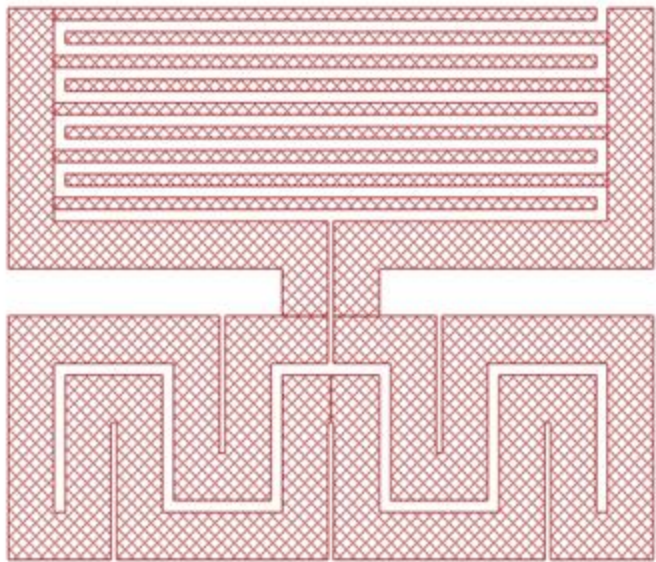


Interdigitated capacitor and thin meandering symmetric coplanar inductor

- When energy (phonons, dark matter, etc) is absorbed, cooper pairs break and create quasiparticles

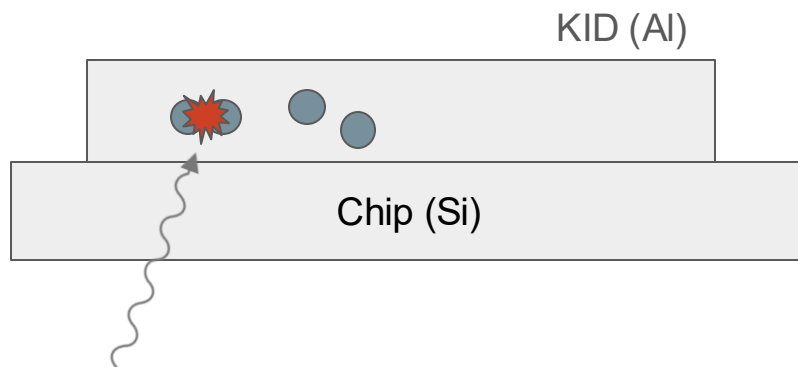


0. Kinetic Inductance Detector

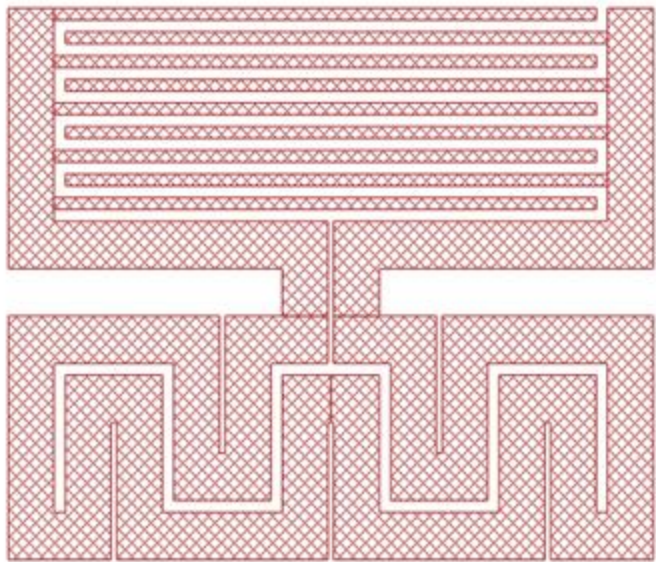


Interdigitated capacitor and thin meandering symmetric coplanar inductor

- Increase in quasiparticle density changes the surface impedance, shifting resonant frequency and quality factor



0. Kinetic Inductance Detector



MKIDs have a combination of kinetic and magnetic inductance

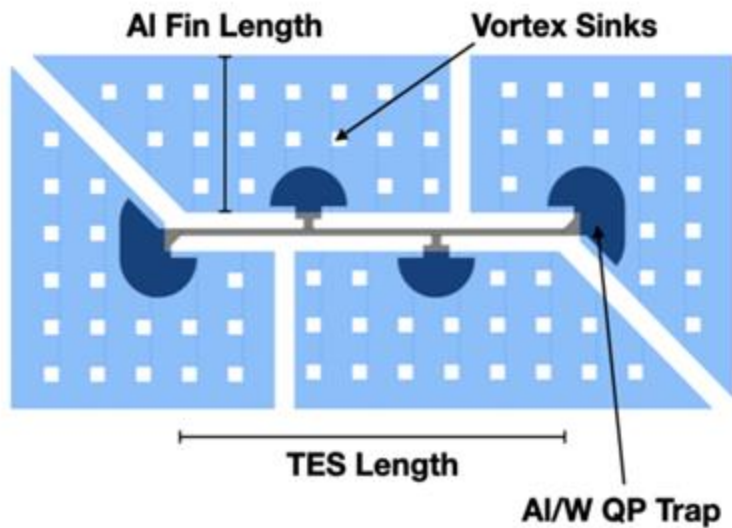
$$L_{\text{total}} = L_K + L_M$$

$$\alpha = \frac{L_K}{L_{\text{total}}} \quad \text{Kinetic-inductance fraction}$$

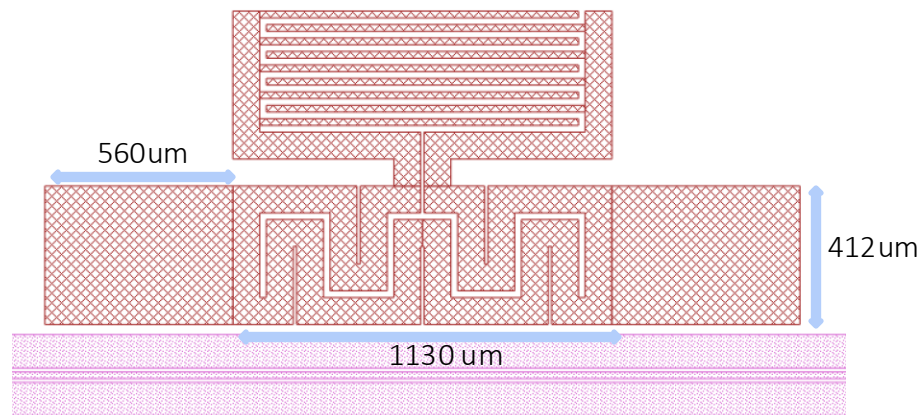
$$f_r = \frac{\omega_r}{2\pi} = \frac{1}{2\pi\sqrt{L_{\text{total}}C}} \propto \frac{1}{\sqrt{L_K + L_M}}$$

Resonant frequency follows definition of LC resonator, will depend on combined inductance

1. Adding Quasiparticle trapping fins



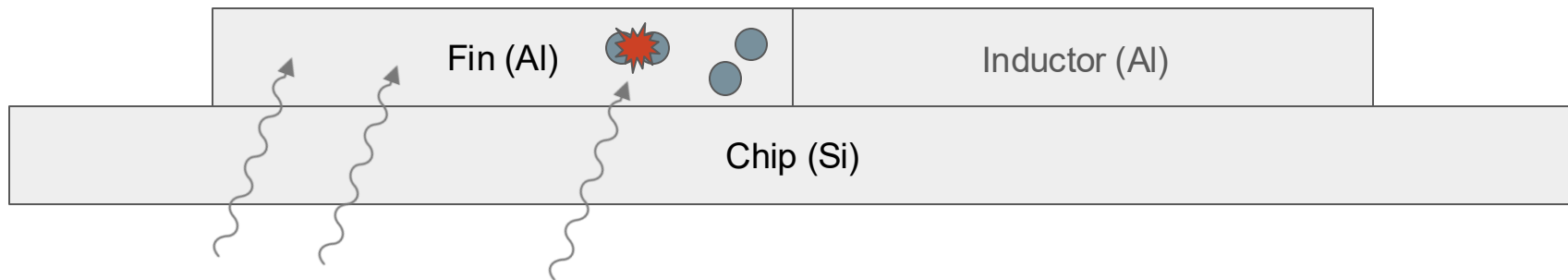
Transition Edge Sensors



KIDs

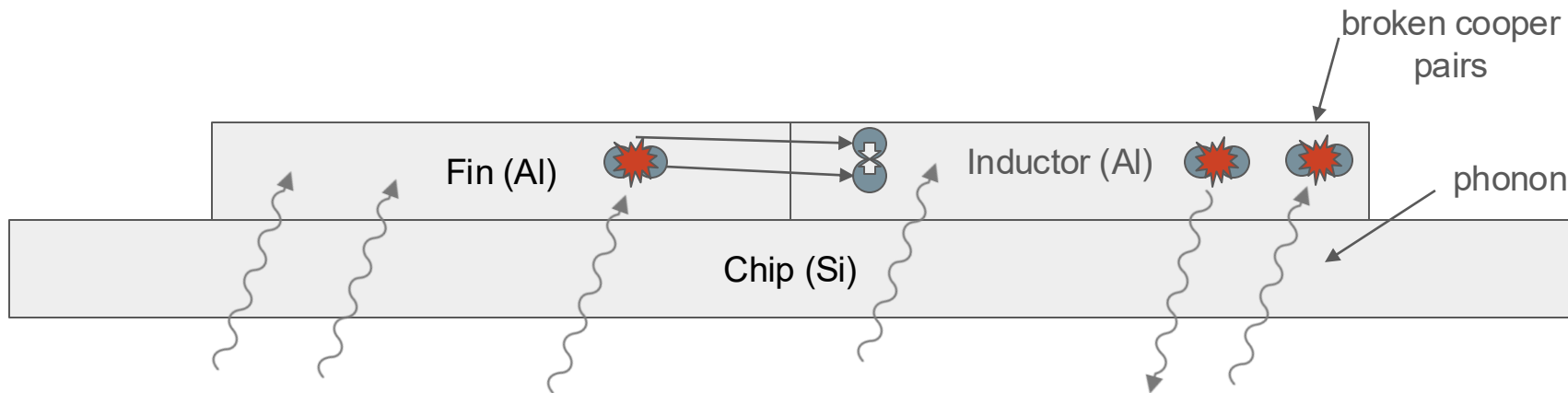
1. Adding Quasiparticle trapping fins

- Phonons transmitted from the chip break Cooper pairs in the fin (Nb) creating quasiparticles
- The quasiparticles diffuse about the fins
- Gap energy of aluminum: 0.34meV, band gap of silicon: 1.1eV (no cooper pairs exist in silicon)



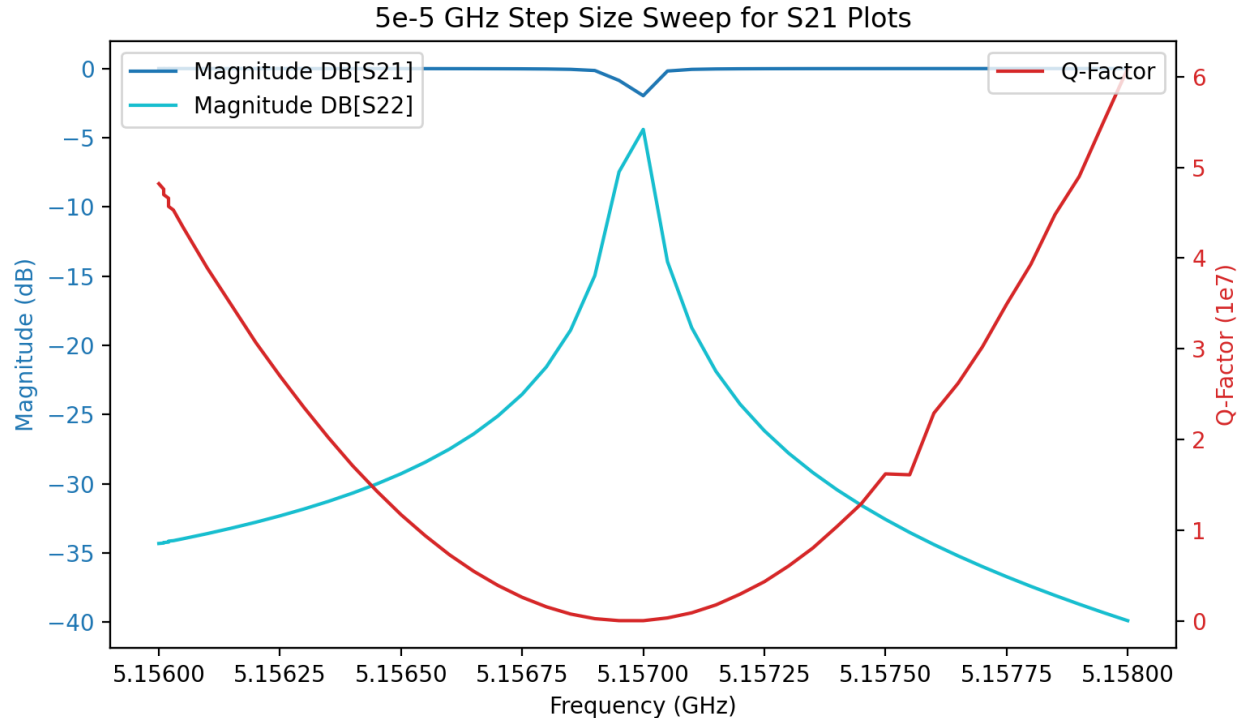
1. Adding Quasiparticle trapping fins

- Some quasiparticles diffuse into the inductor, recombine, and emit phonons
- These phonons then go on and break more cooper pairs in the inductor
- When produced, the inductor will be made of a lower gap material



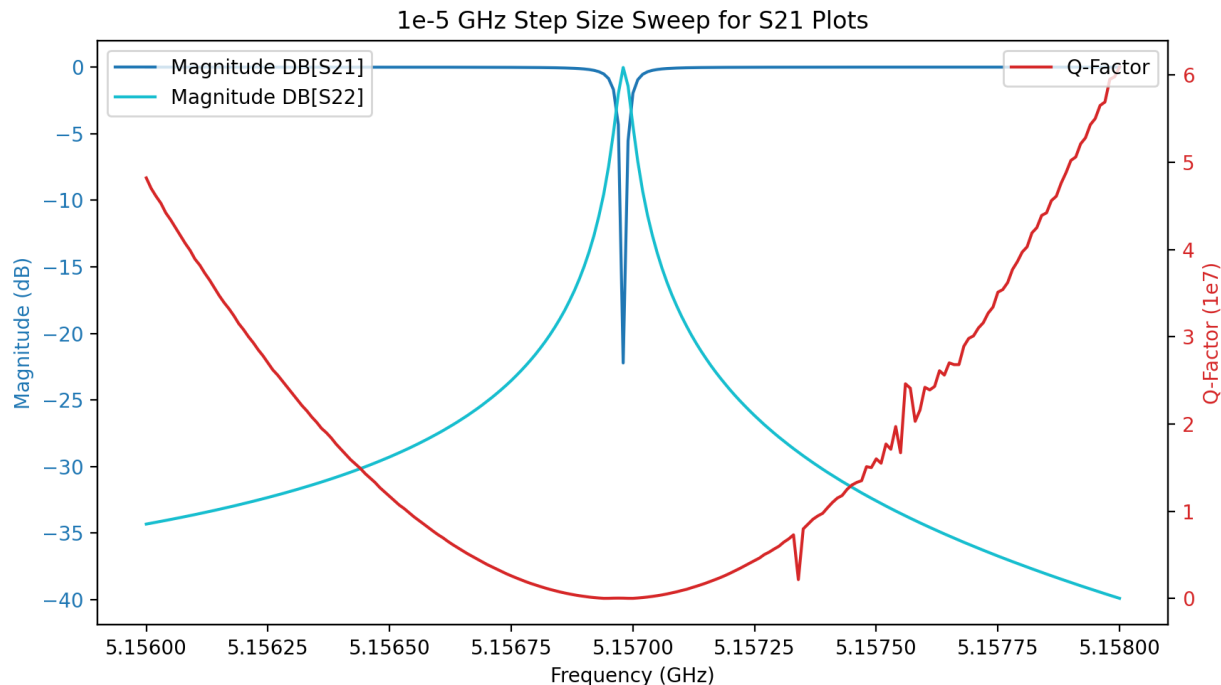
Finding Resonant Frequency and Q factor

- Original 2-port method, finding the peak of S21
- 20-hour long simulations



Finding Resonant Frequency and Q factor

- The step size influenced how precise we could estimate the resonant frequency off the image



Finding Resonant Frequency and Q factor

- 3-port method (Jiansong Gao) reduces computation time to ~20 minutes/sweep
 - only 2 data points required

$$Z_{in,3} = j\omega L + \frac{1}{j\omega C} + R \quad \text{Standard expression for impedance of RLC series circuit}$$

- At the resonance frequency, $\text{Im}[Z_{in,3}]$ crosses 0
 - The first two terms on the right side are imaginary components of the impedance, so

$$\cancel{Z_{in,3} = j\omega L + \frac{1}{j\omega C} + R} \quad \longrightarrow \quad Z_{in,3} = R \quad \text{at resonance frequency}$$

$$Z_{in,3} = j\omega L - j\omega C + R$$

Finding Resonant Frequency and Q factor

$$Z_{in,3} = R \left[1 + 2jQ \left(\frac{f - f_r}{f_r} \right) \right] \quad \text{Equation for impedance near series resonance}$$

If $f = f_r$ then the second term inside bracket equals 0

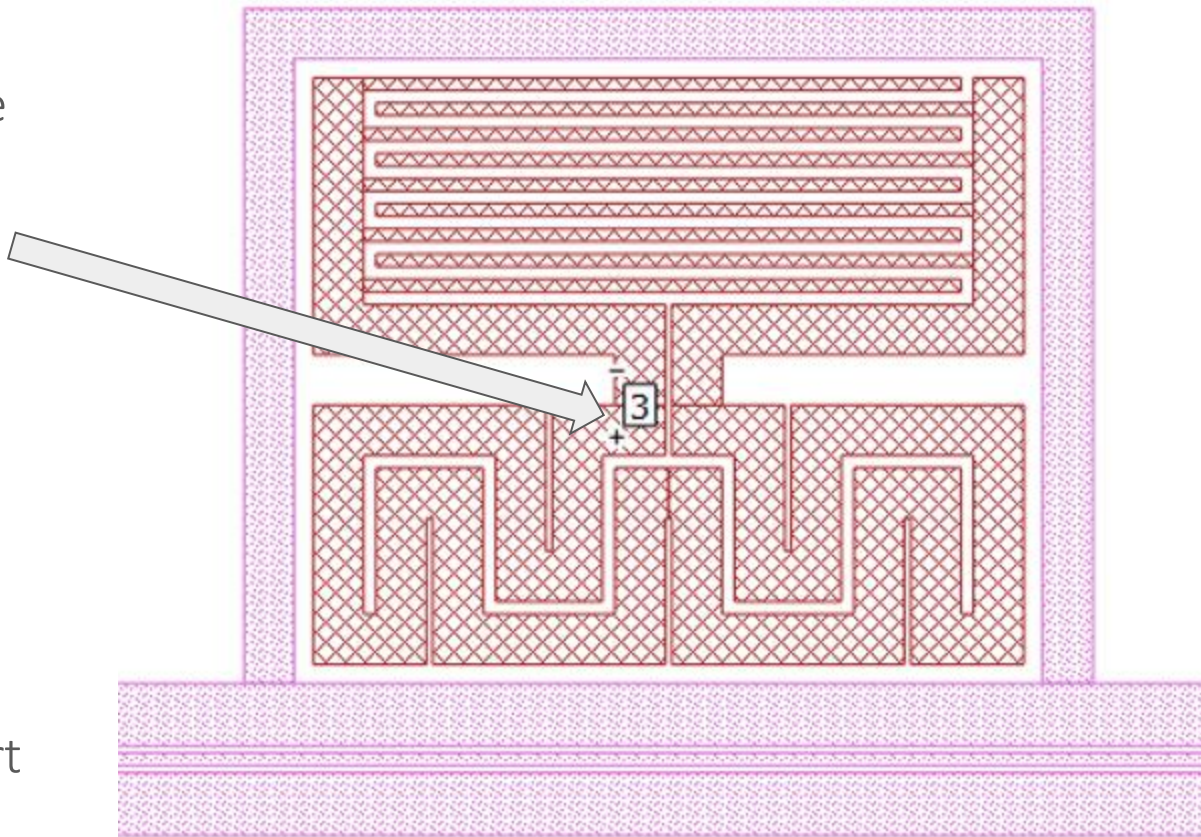
f_r Resonance frequency is the frequency at which $\text{Im}[Z_{in,3}] = 0$

R Value of $\text{Re}[Z_{in,3}]$ at resonance frequency

k Slope of $\text{Im}[Z_{in,3}]$

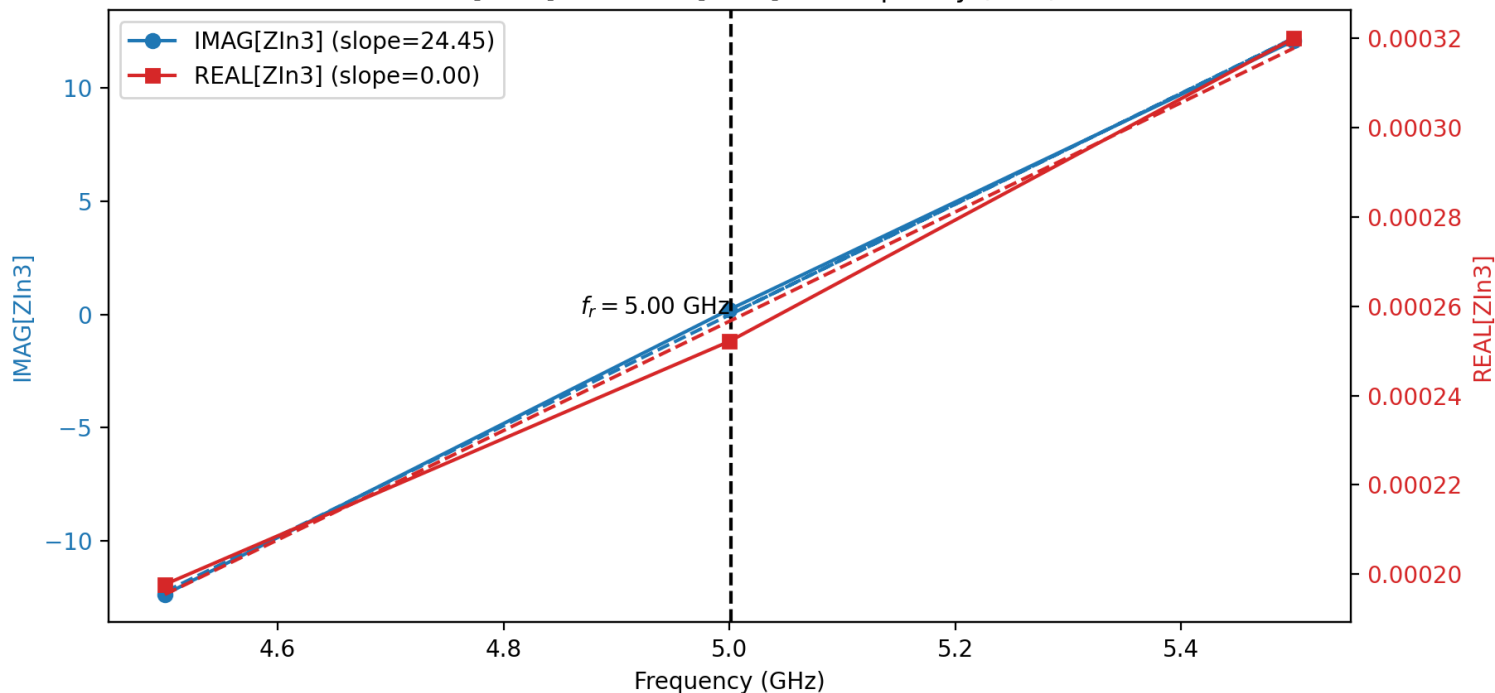
$$Q = \frac{k f_r}{2R} \quad \text{Quality factor of the kinetic inductance detector (given in paper)}$$

3rd 'internal' port inside
the resonator
– added where the IDC
joins meandering
inductor

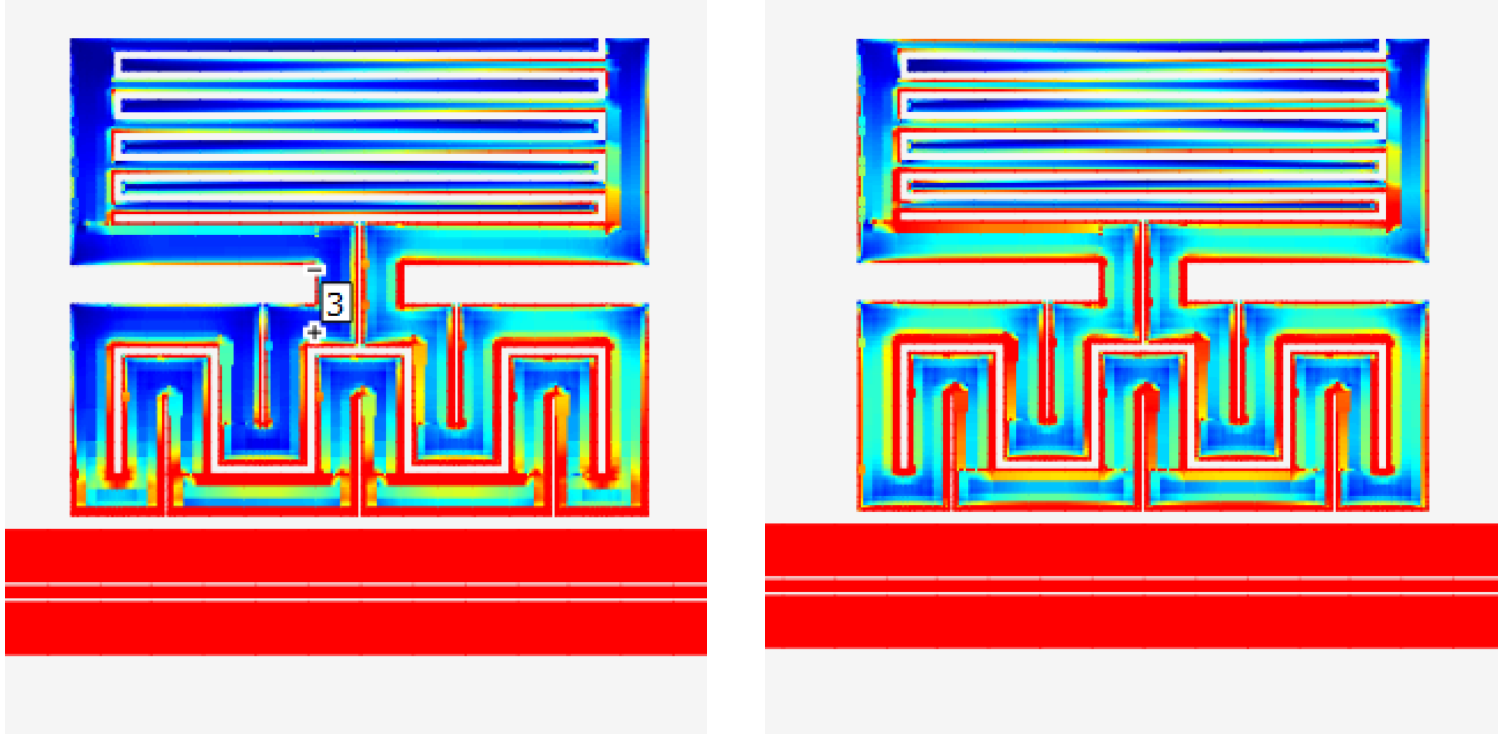


The 3-port method
measures the input
impedance of the port

IMAG[ZIn3] and REAL[ZIn3] vs Frequency (GHz)



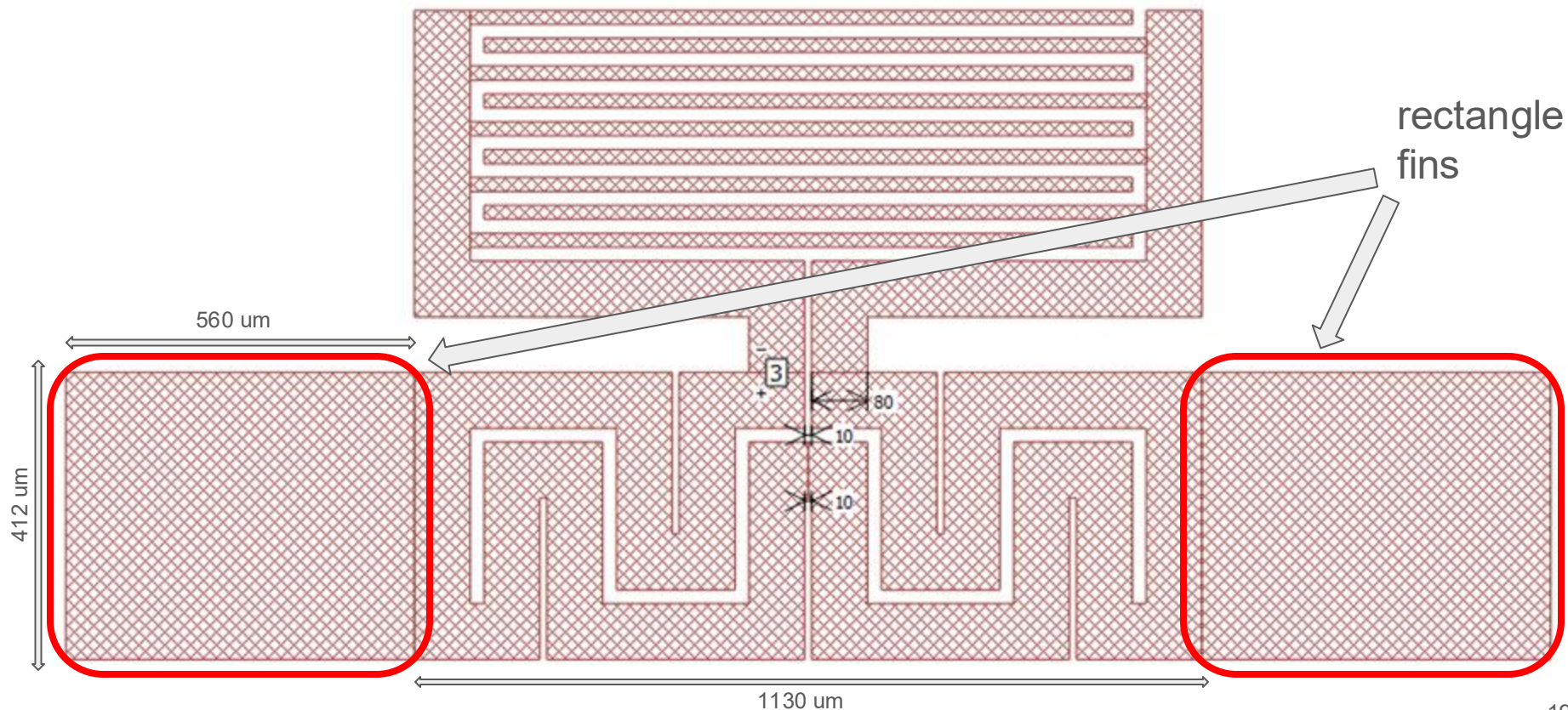
Using the 3-port method, 2 datapoints would be sufficient



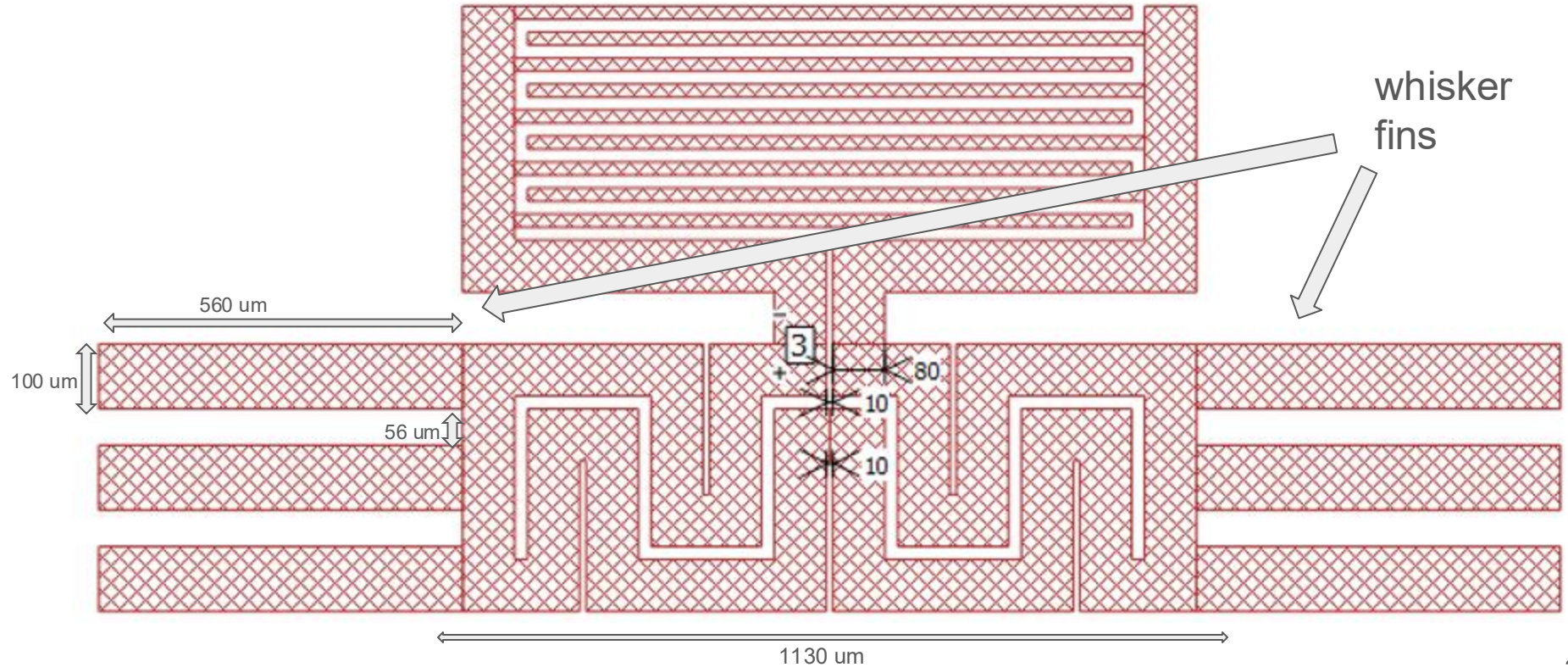
current density generated at resonant frequency without 3-port

- extra port generated asymmetrical current densities

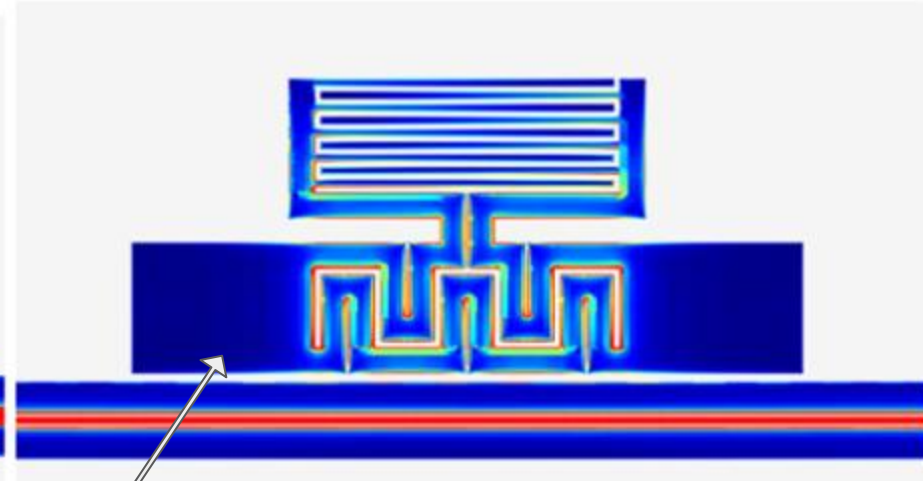
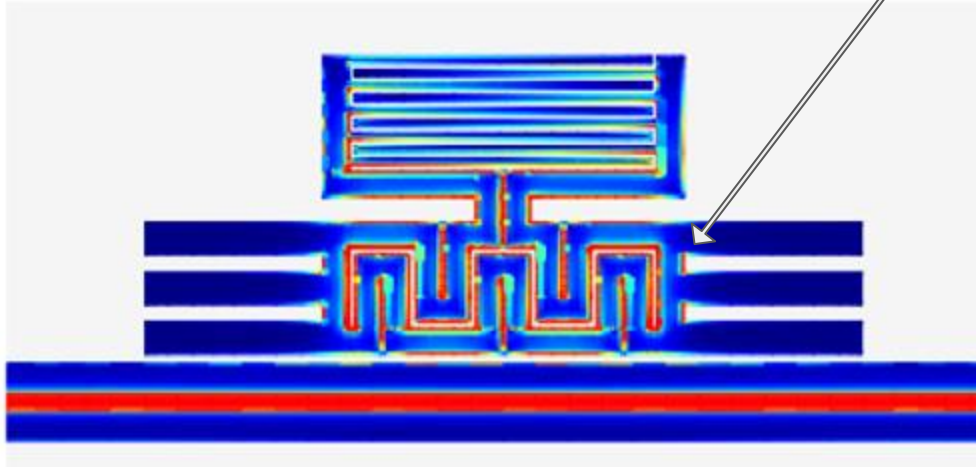
1st Design - Rectangle Fins



1st Design - Whisker Fins



high current density on the fins



Fin width 560 um

Changing Circuit Geometry–Decreasing Inductor Volume

$$n_{cp} = \frac{N_{cp}}{Volume}$$

Because we're sensitive to quasiparticle density, decreasing the volume of the resonator increases phonon collection efficiency

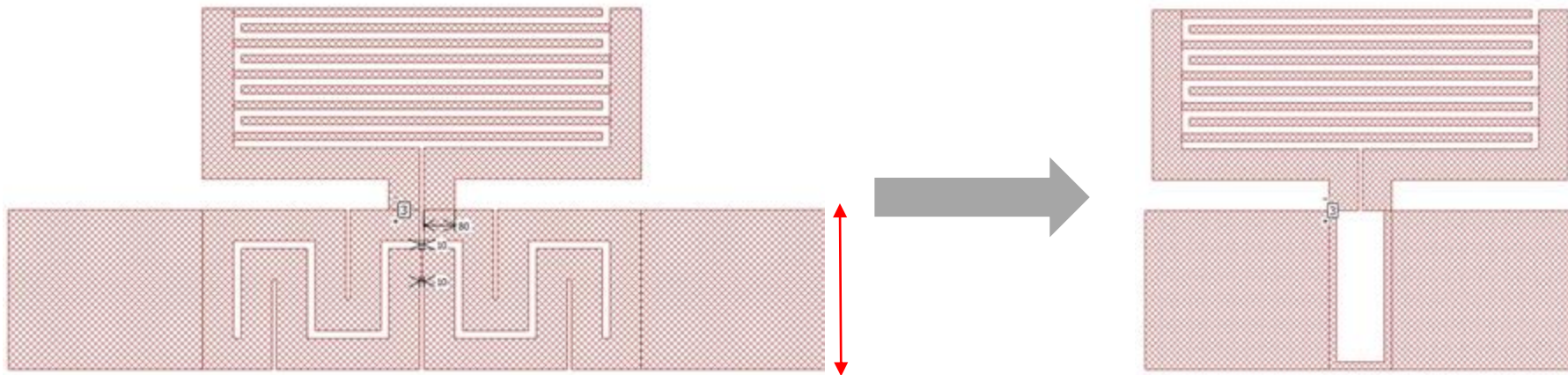
$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$5 \leq f_r \leq 6$ (GHz) for KI-TWPA, so inductance can't change

$$L = L_{sq} \frac{length}{width}$$

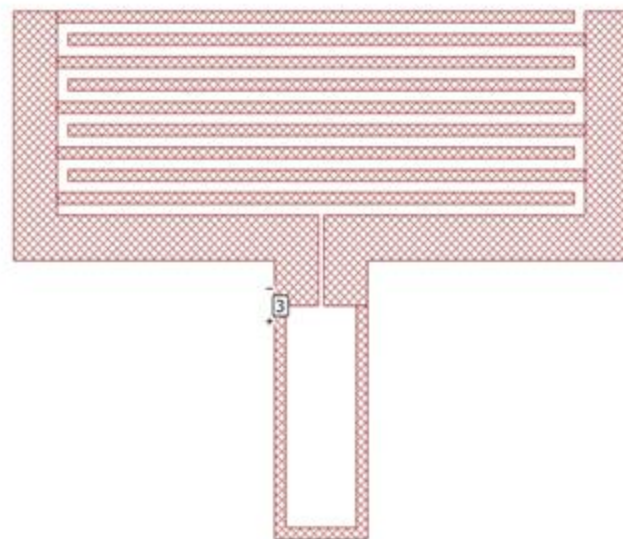
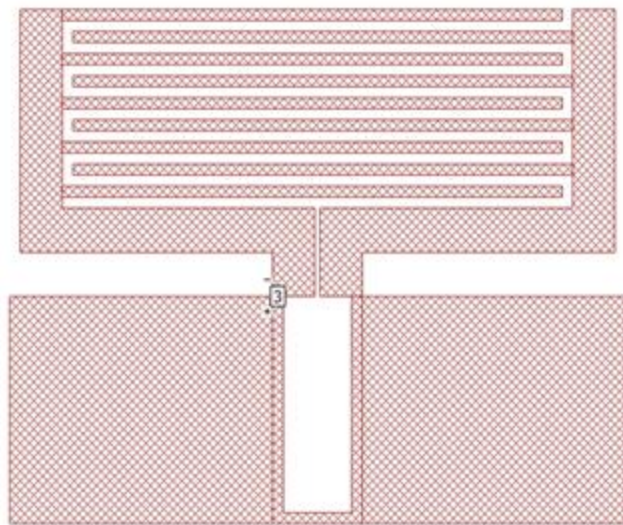
Change length and width by same factor

Design Choices:



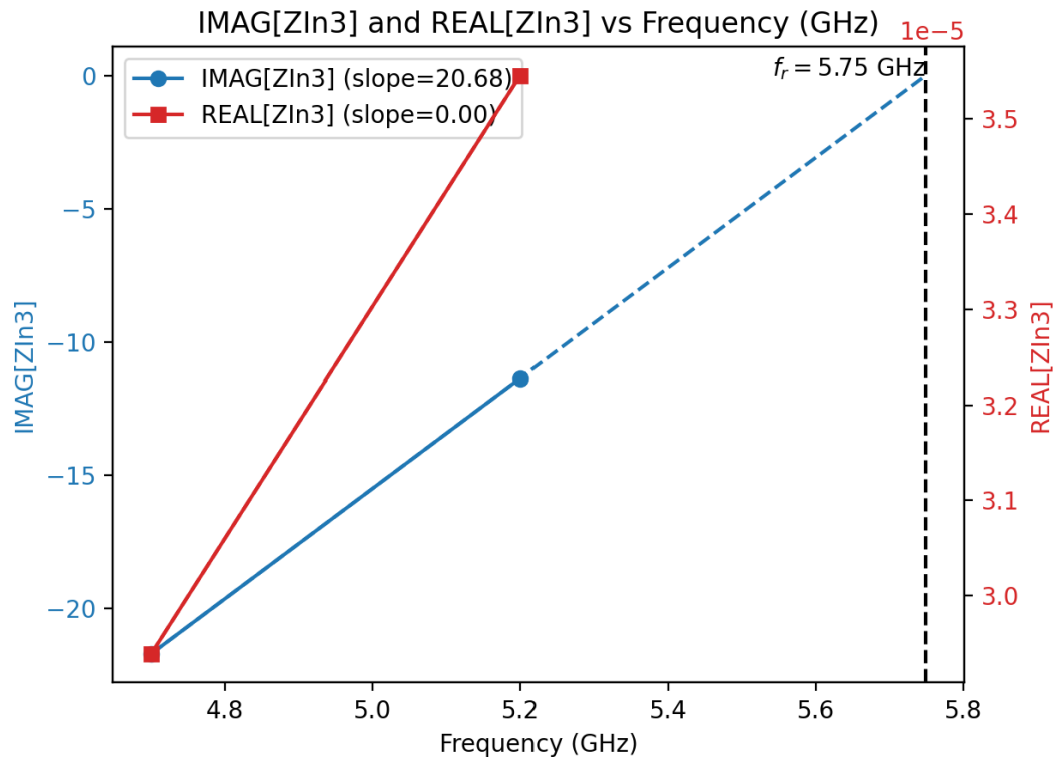
- Only constraint in design: **fin height (412 um)**
 - To maintain same fin size and the touching border of fin and inductor
- U-shape, decreased length and width by same factor
 - Maximum factor possible = 5.18511

Design Choices: With or Without Fins

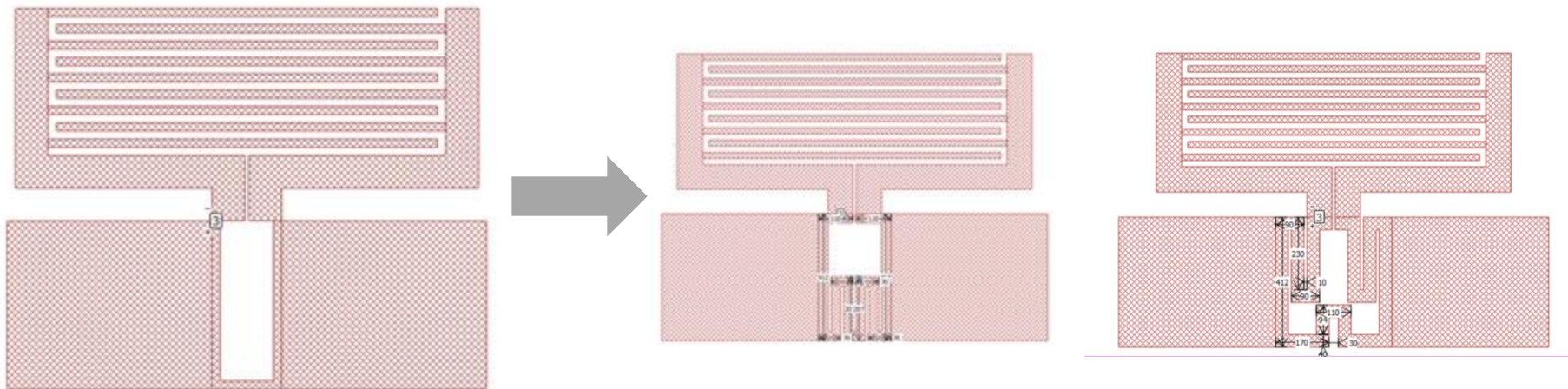


- Goal: Determine the effect of fins on resonant frequency

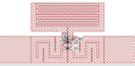
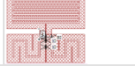
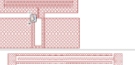
Extracting Resonant Frequency



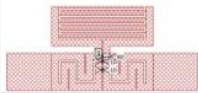
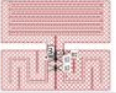
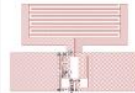
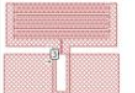
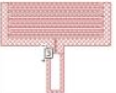
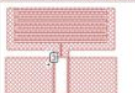
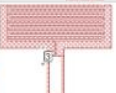
Design Choices:



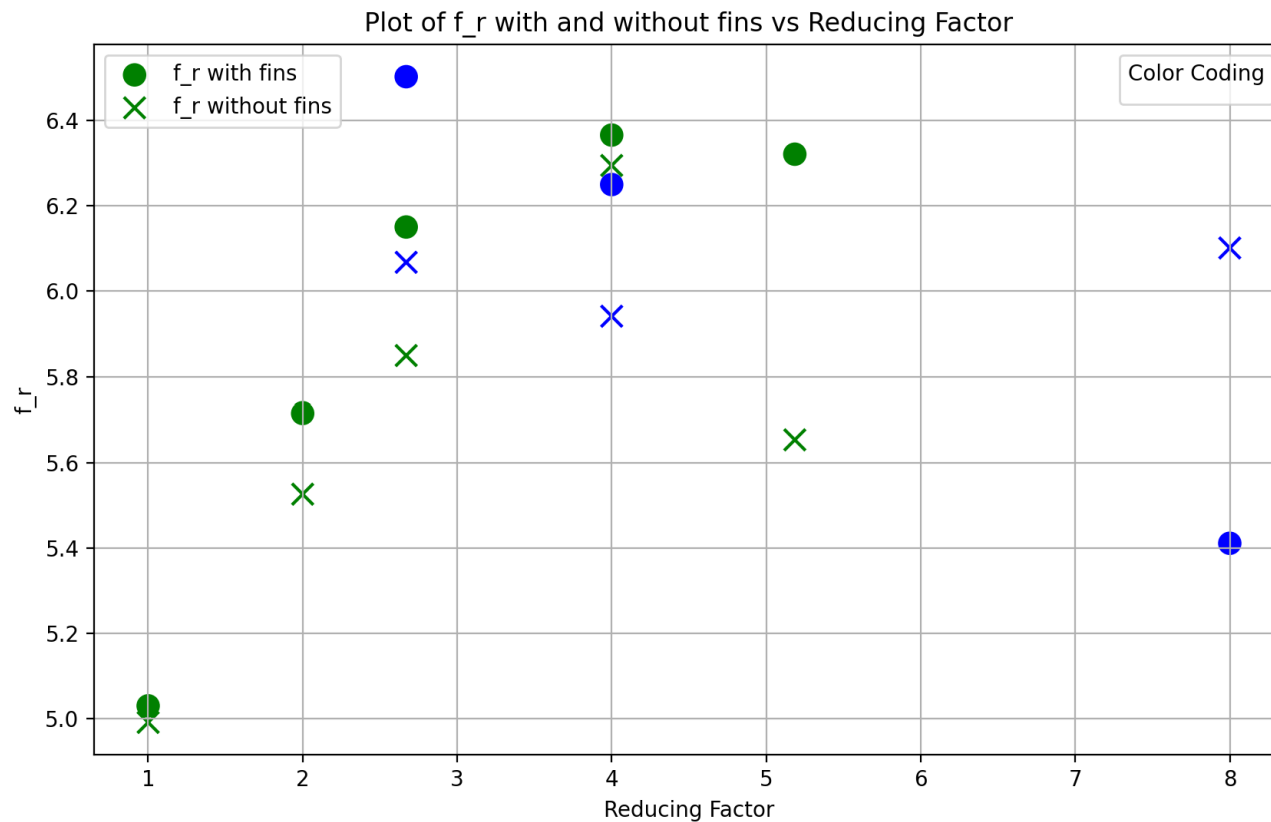
- U-shape: not an inductor but was used to test for resonant frequency changes
- Added meanders:
 - 1. length/width reduced by same factor means the number of least squares is maintained and the inductance shouldn't change
 - 2. Arbitrary choice of length to maintain meanders

reducing factor	with vs without fin: ideal vs actual		with fin	ideal f _r with fins	f _r with fins	<-- Q factor	without fin	f _r without fins	<-- Q factor	width	length	same factor?
1	-0.78%	0.00%		5.029983278	5.029983278	166680.298		4.9911503	239077.9808	80	4994	yes, meander
2	-3.42%	-13.63%		5.029983278	5.715338207	312433.5415		5.526127146	774084.2782	40	2497	yes, meander
2.67	-5.15%	-25.67%		5.029983278	6.15061005	496289.6515		5.849334895	2182028.476	30	1870	yes, meander
2.67	-7.17%	8.64%		7.117776174	6.502996563	442325.9798		6.06788065	1551713.317	30	934	no, u-shape
4	-1.13%	-26.54%		5.030990786	6.366102717	453380.8883		6.294863516	711220.2757	20	1243	yes, meander
4	3.24%	0.09%		5.754222187	5.749212751	1411381.58		5.9414692	3691387.035	20	954	no, u-shape
5.18511	-11.81%	-25.67%		5.029983444	6.321257671	109035.7034		5.6535054	1068341.11	15	963	yes, u-shape
8	11.32%	-34.37%		4.026858214	5.411062622	1179683.371		6.10146072	265573.5293	10	974	no, u-shape

- Resonant frequency remains in 5-6 GHz range
 - *without fins* and *changed inductance* design raised resonant frequency to above 6 GHz

reducing factor	with vs without fin	with fin	f_r with fins	without fin	f_r without fins	same factor?
1	-0.78%		5.029983278		4.9911503	yes, meanders
2	-3.42%		5.715338207		5.526127146	yes, meanders
2.67	-5.15%		6.15061005		5.849334895	yes, meanders
2.67	-7.17%		6.502996563		6.067880651	no, u-shape
4	-1.13%		6.366102717		6.294863516	yes, meanders
4	3.24%		5.749212751		5.94146921	no, u-shape
5.18511	-11.81%		6.321257671		5.6535054	yes, u-shape
8	11.32%		5.411062622		6.10146072	no, u-shape

- Resonant frequency remains in 5-6 GHz range



Changing Circuit Geometry–Increasing Capacitor Volume

$$n_{cp} = \frac{N_{cp}}{Volume}$$

Decreasing the volume of the resonator increases phonon collection efficiency

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

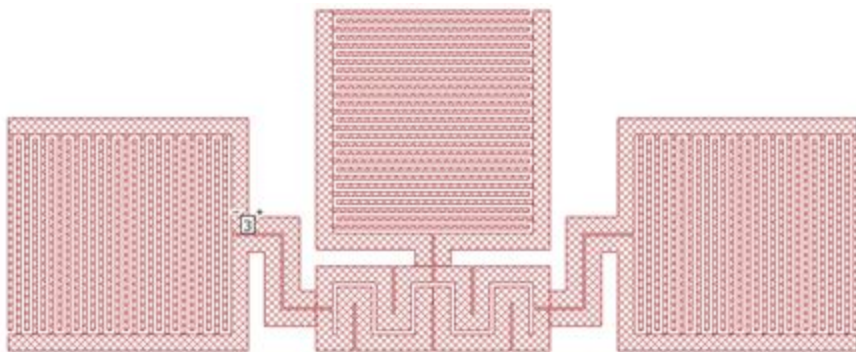
$5 \leq f_r \leq 6$ (GHz) for KI-TWPA, so inductance can't change

$$\frac{1}{C_{total}} = \frac{1}{C_1} + \frac{1}{C_2}$$

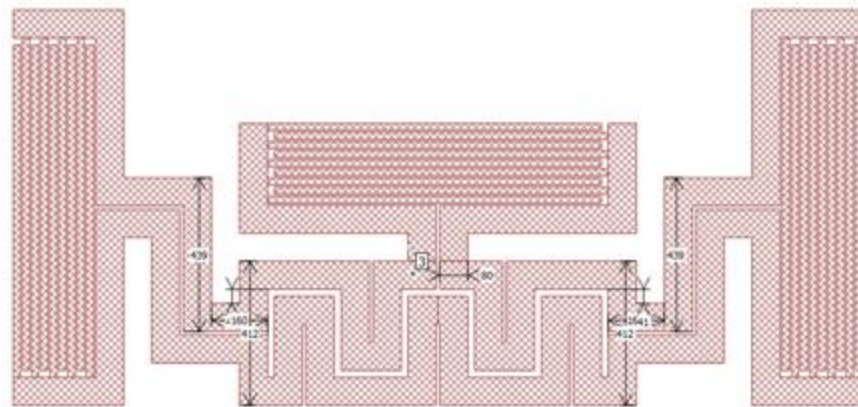
In series

$$C_{total} = C_1 + C_2$$

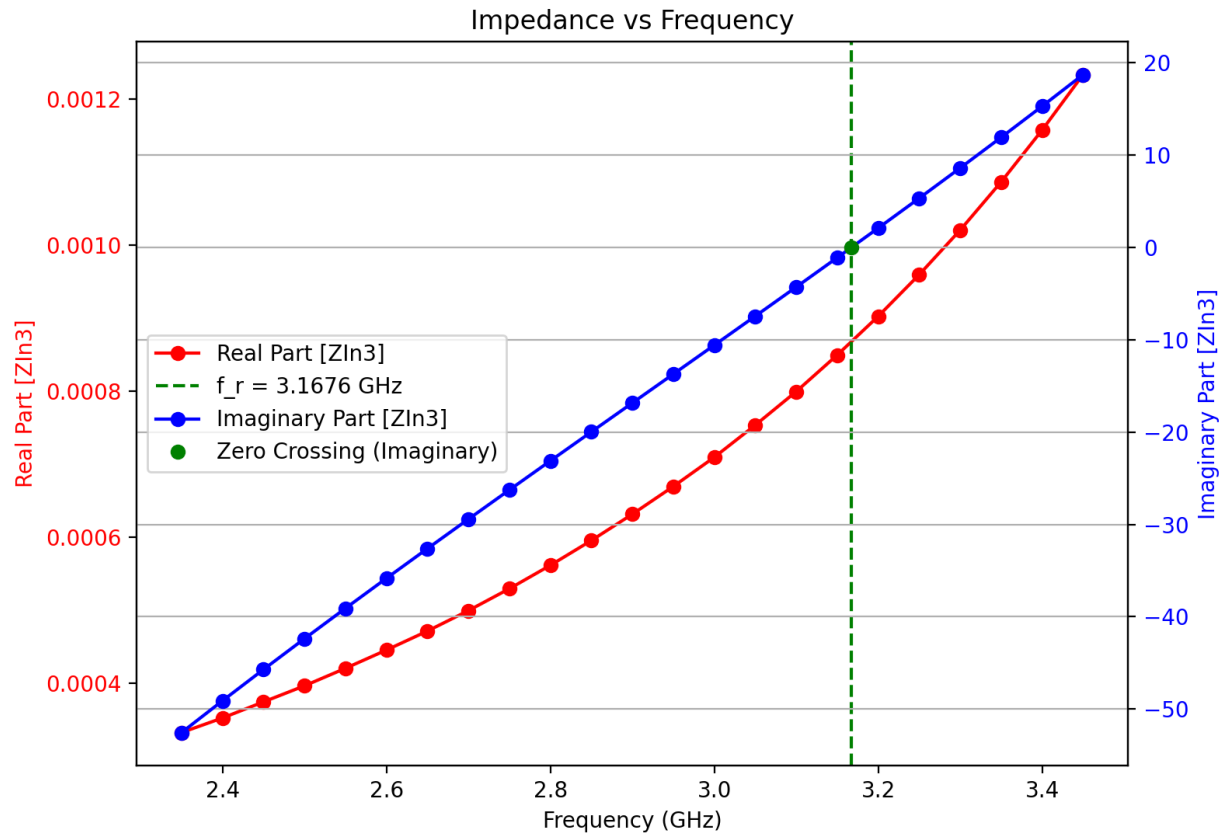
In parallel



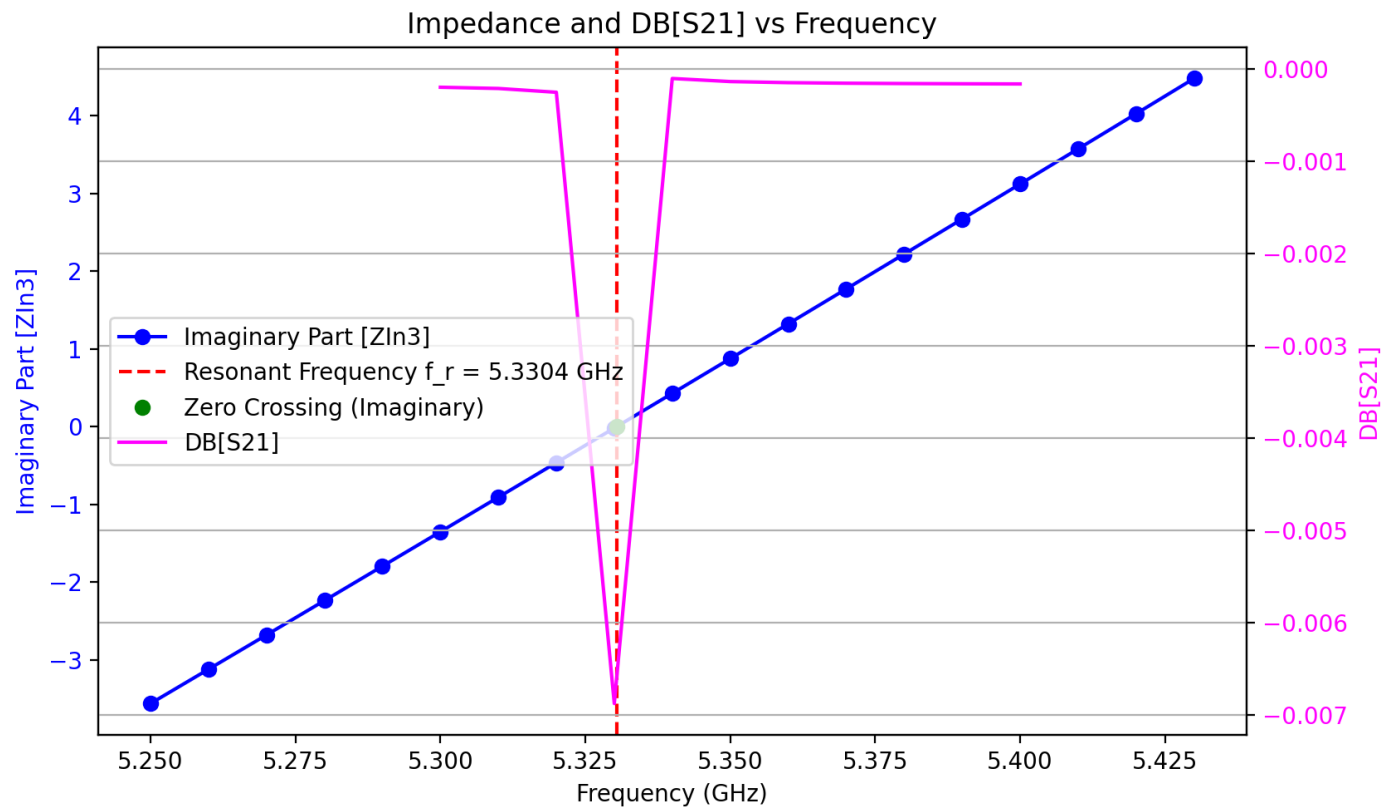
- Tripling the number of inter-digited capacitor teeth and adding 2 more fins
 - $9 \rightarrow 27$
- Resonant frequency: 3.18 GHz

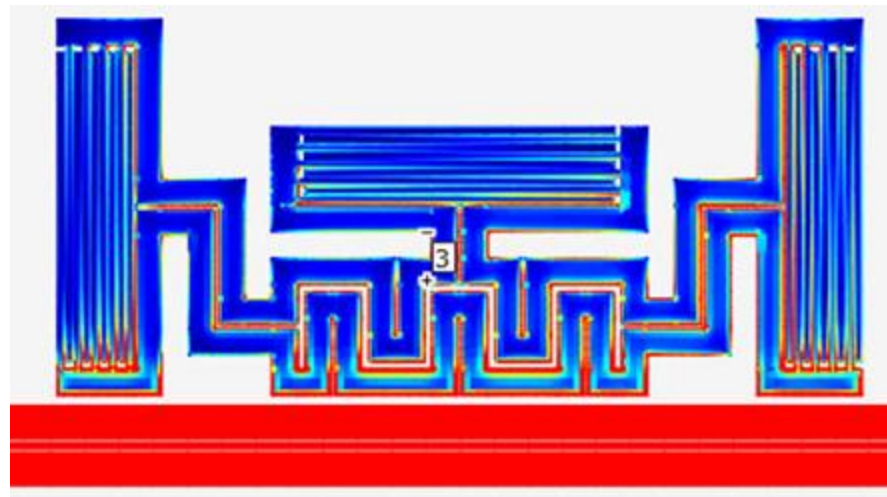
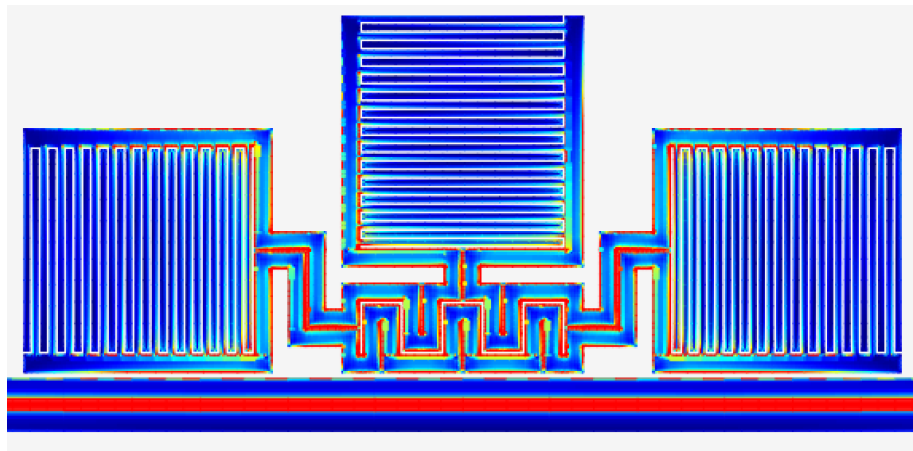


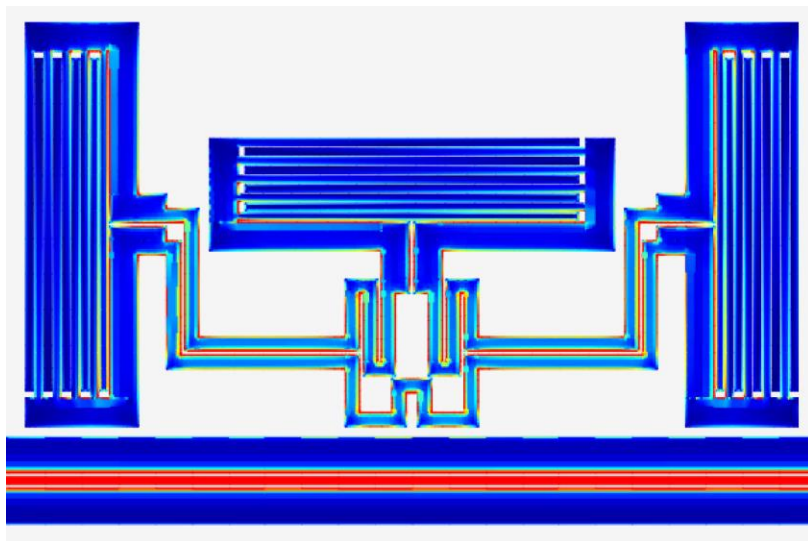
- Decreasing the distance between inter-digited capacitor teeth by a factor of 3
 - 20um to 6um
- Resonant frequency: 5.3 GHz



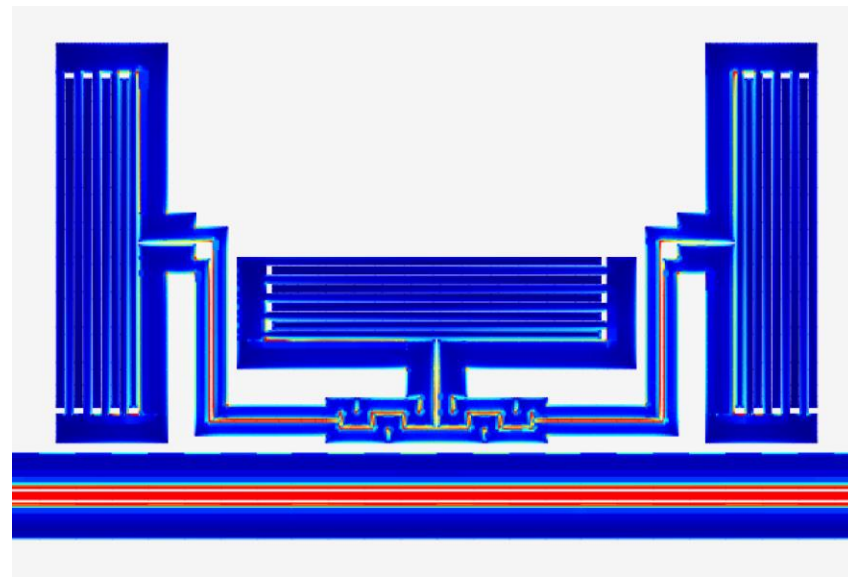
- Real component of impedance is nonlinear
- Step size impacts calculated Q factor
- Raises the question if the 3 port method is still valid







- Decreasing the meandering inductor length by 2
- Resonant frequency: 5.238 GHz



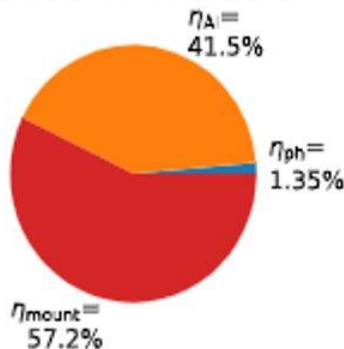
- Decreasing length of meandering inductor by 2
- Resonant frequency: 5.563 GHz

2. Redesigning KID box mount–Motivations

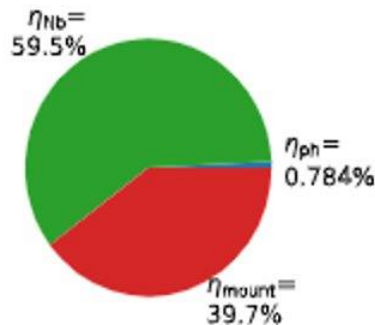
$$E_{\text{abs}} = \eta_{\text{ph}} E$$

Comparing the energy a substrate deposits to the energy the resonator absorbs

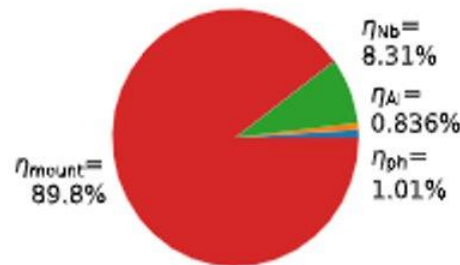
DMLE2, one resonator



OW200127

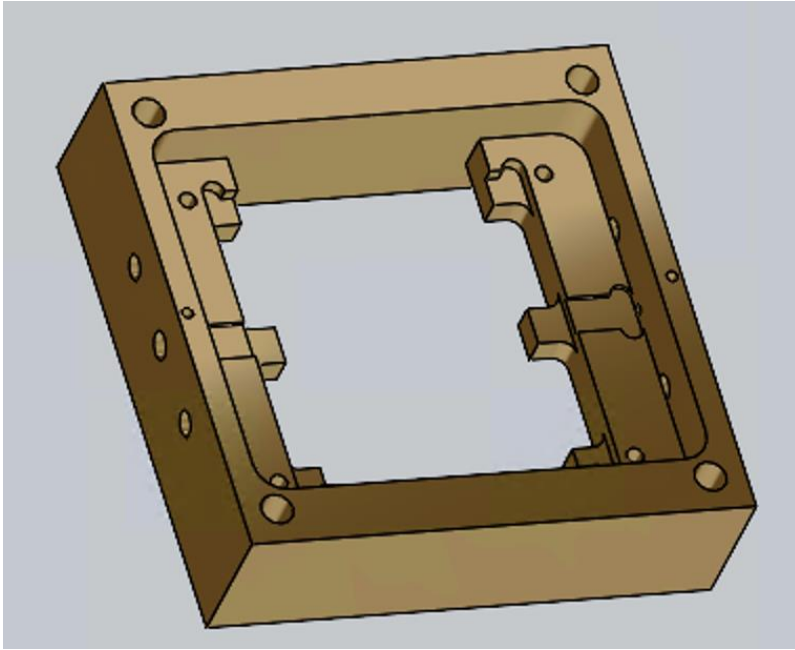


B240103



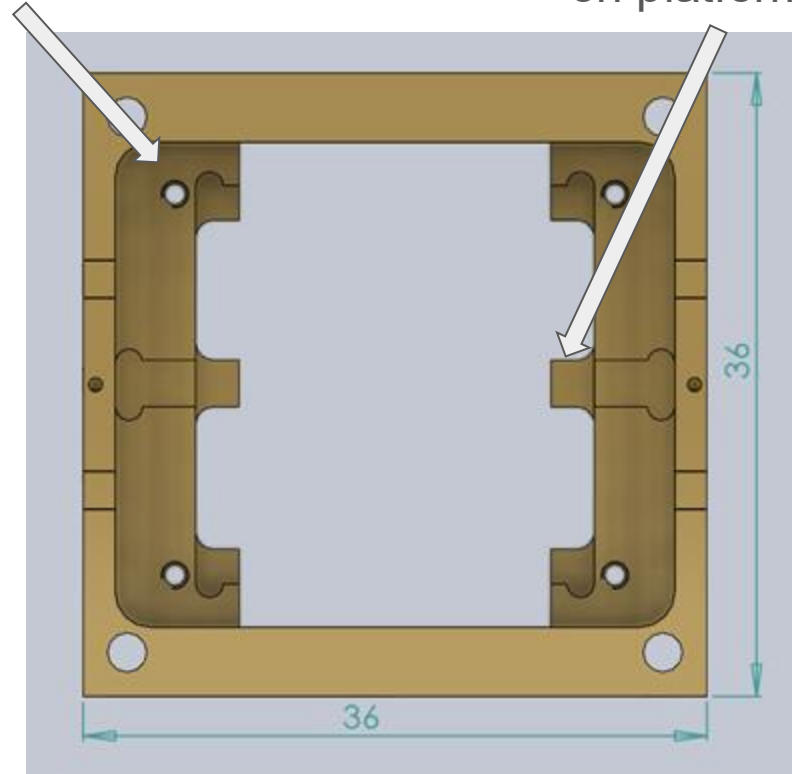
- Most phonon energy loss is due to **mounting**

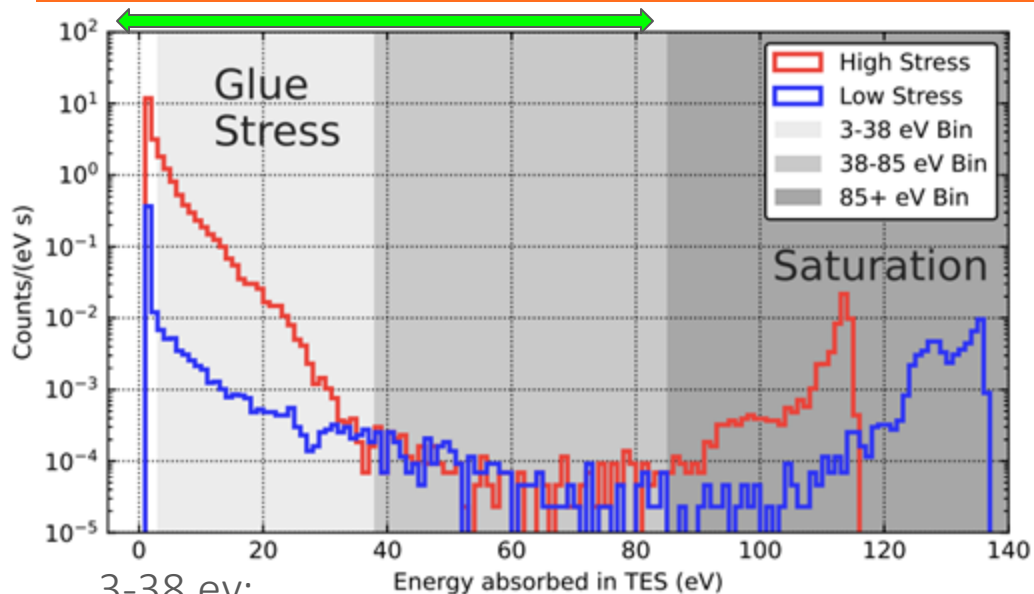
Original Design



O-ring clamp

KID resting
on platform





Glue results in high mounting stress (green region) at low energy depositions

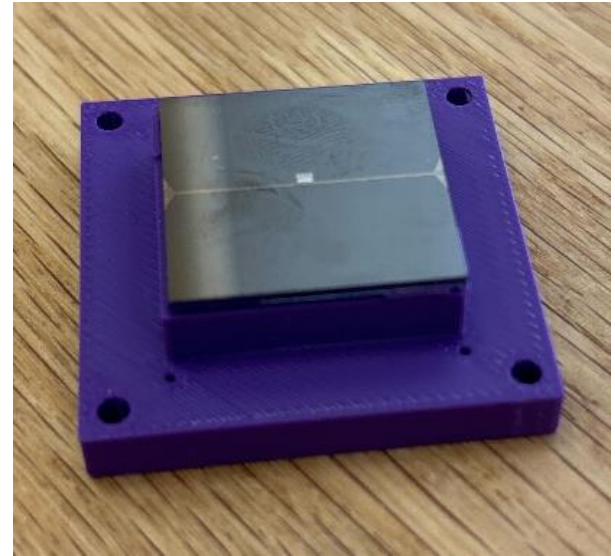
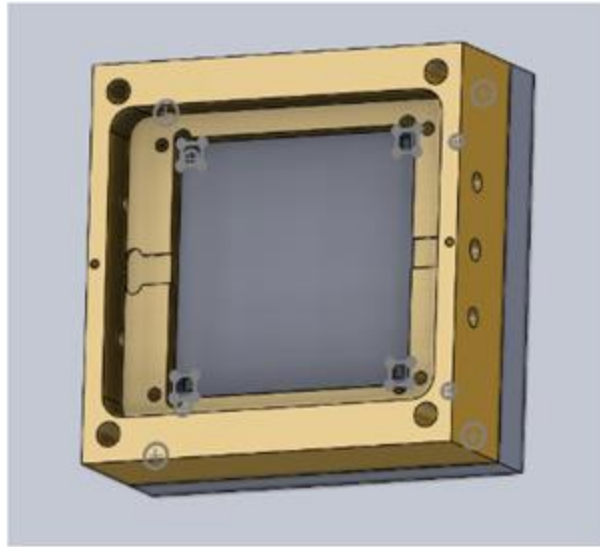
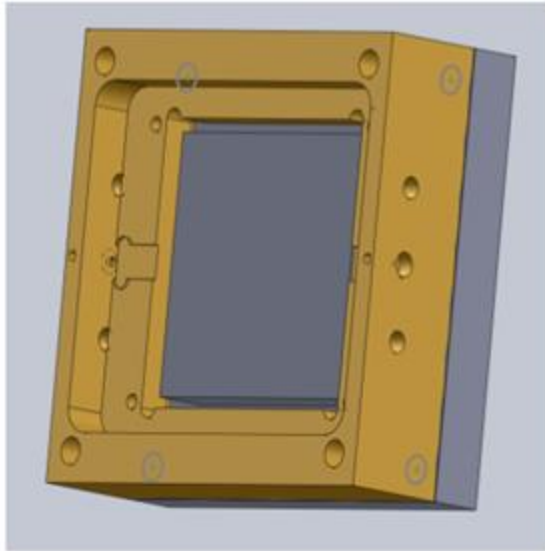
3-38 eV:

- 2-3 mag. diff. due to mounting stress
- (GE varnish contraction relative to HS crystal $\rightarrow$ stress relaxes through microfracture $\rightarrow$ eV scale broken bonds $\rightarrow$ phonon energy released)

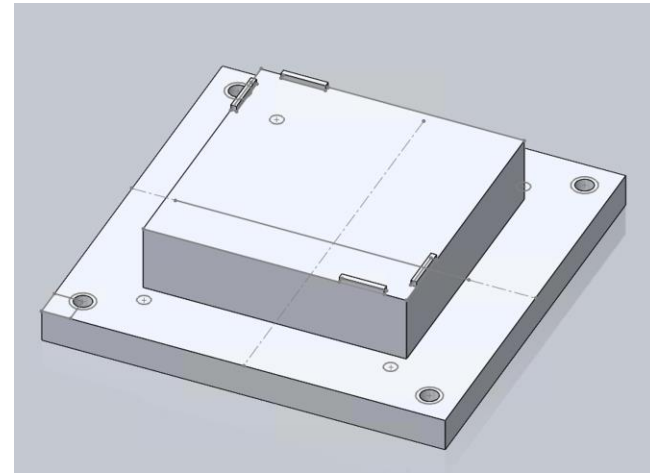
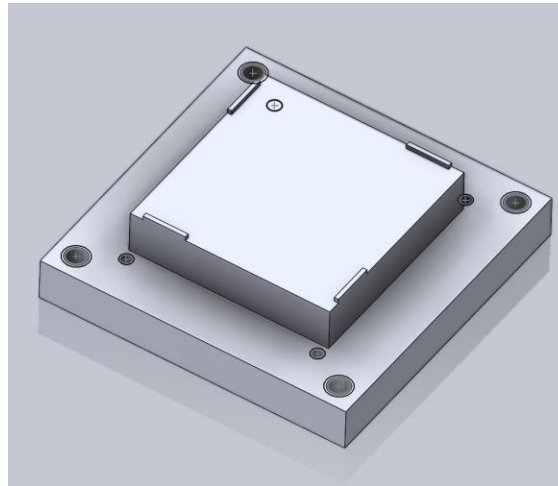
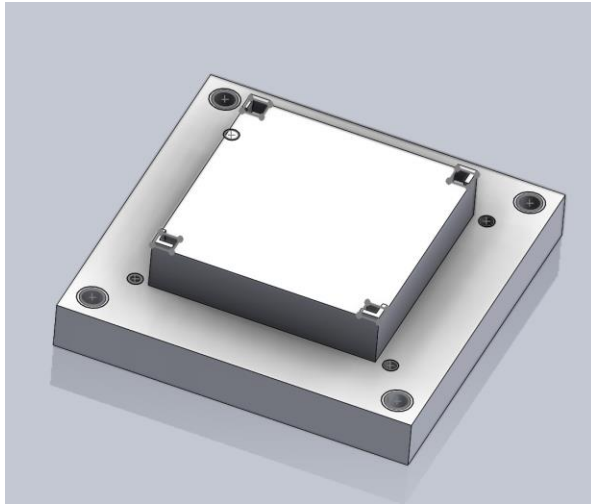
38-86 eV:

- due to stress in QET thin films
- (from thermal contraction during cooldown & film deposition)

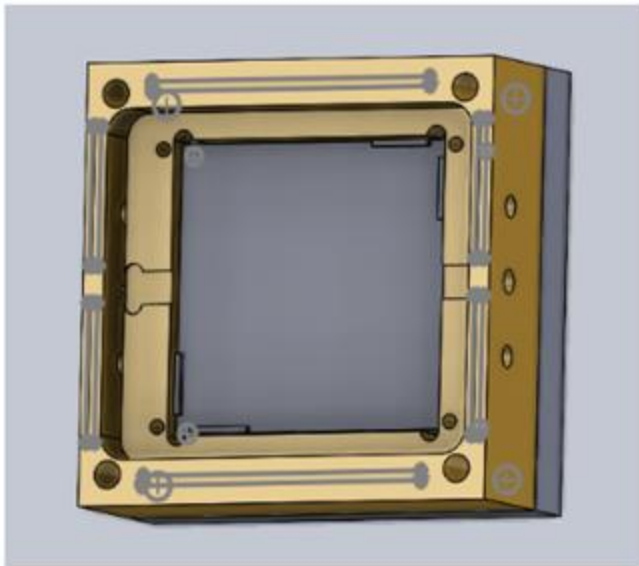
Design Iterations



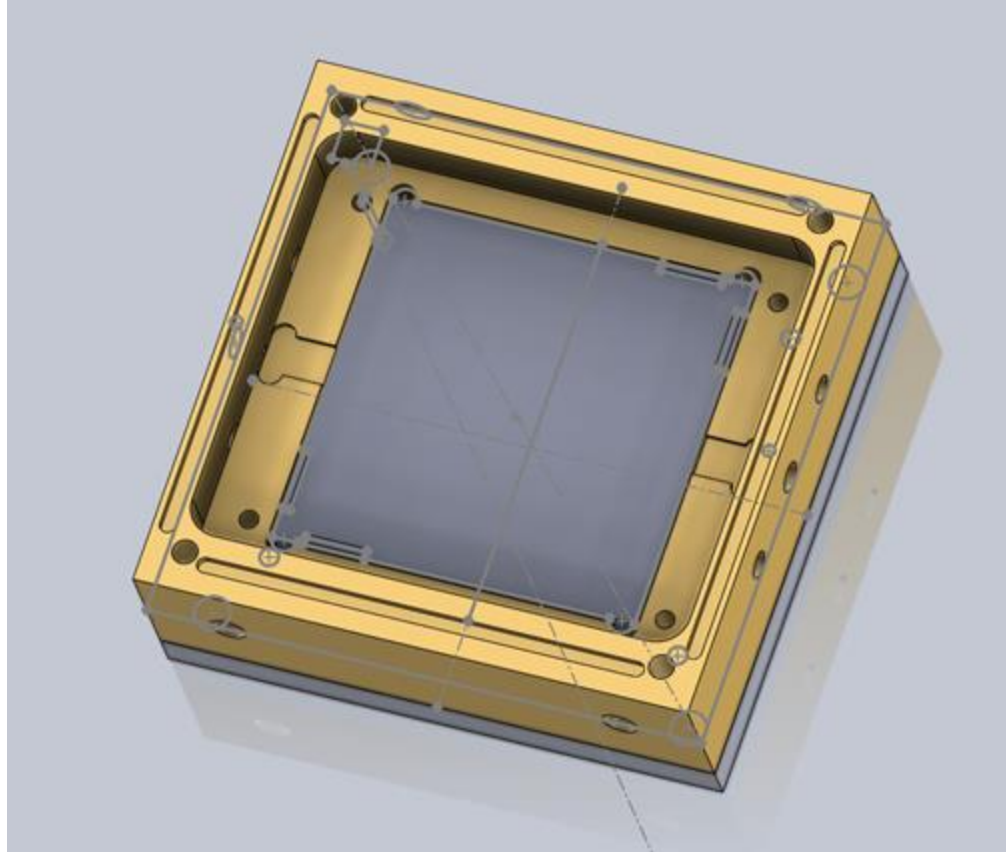
Design Iterations



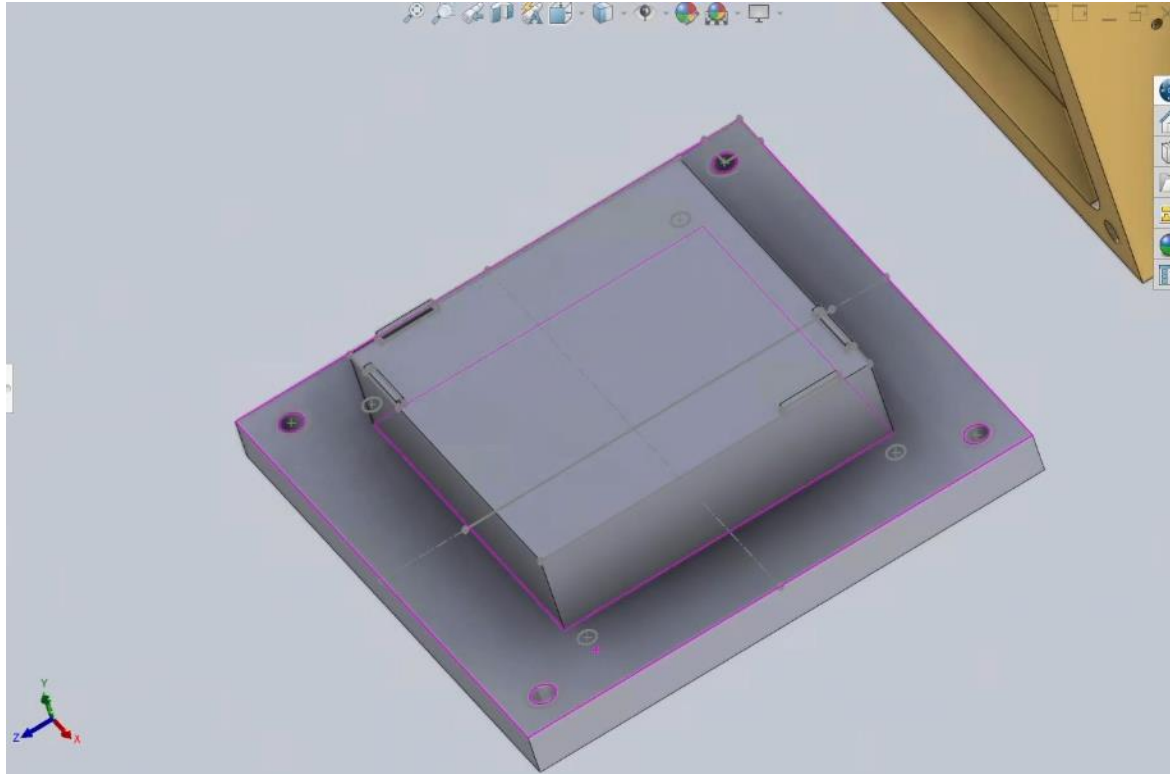
3D Printed Trials



Final Design



Video Demonstration



Next Steps

- Machine the box
- Test suspended wire bond chip
- Produce chip with modified geometry (fins with higher phonon collection efficiency)

A special thank you to

- Professor Sunil Golwala for giving us the opportunity to work in this lab, supporting our research interests, and providing guidance
- Our mentors Brandon Sandoval and Osmond Wen for their patience in taking the time to work through all the design iterations and physics with us.
- The Golwala Group and ObsCos

Feedback is appreciated!



References

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Thank you for listening!